

Radiation? More like Stay-away-i-ation

Background Radiation

A combination of ambient radiation (radiation that is present in the environment) as well as a specific type: ionized radiation (radiation that occurs because of energy from ionized atoms i.e. alpha, beta, gamma, neutron)

Radiation can occur:

Naturally.

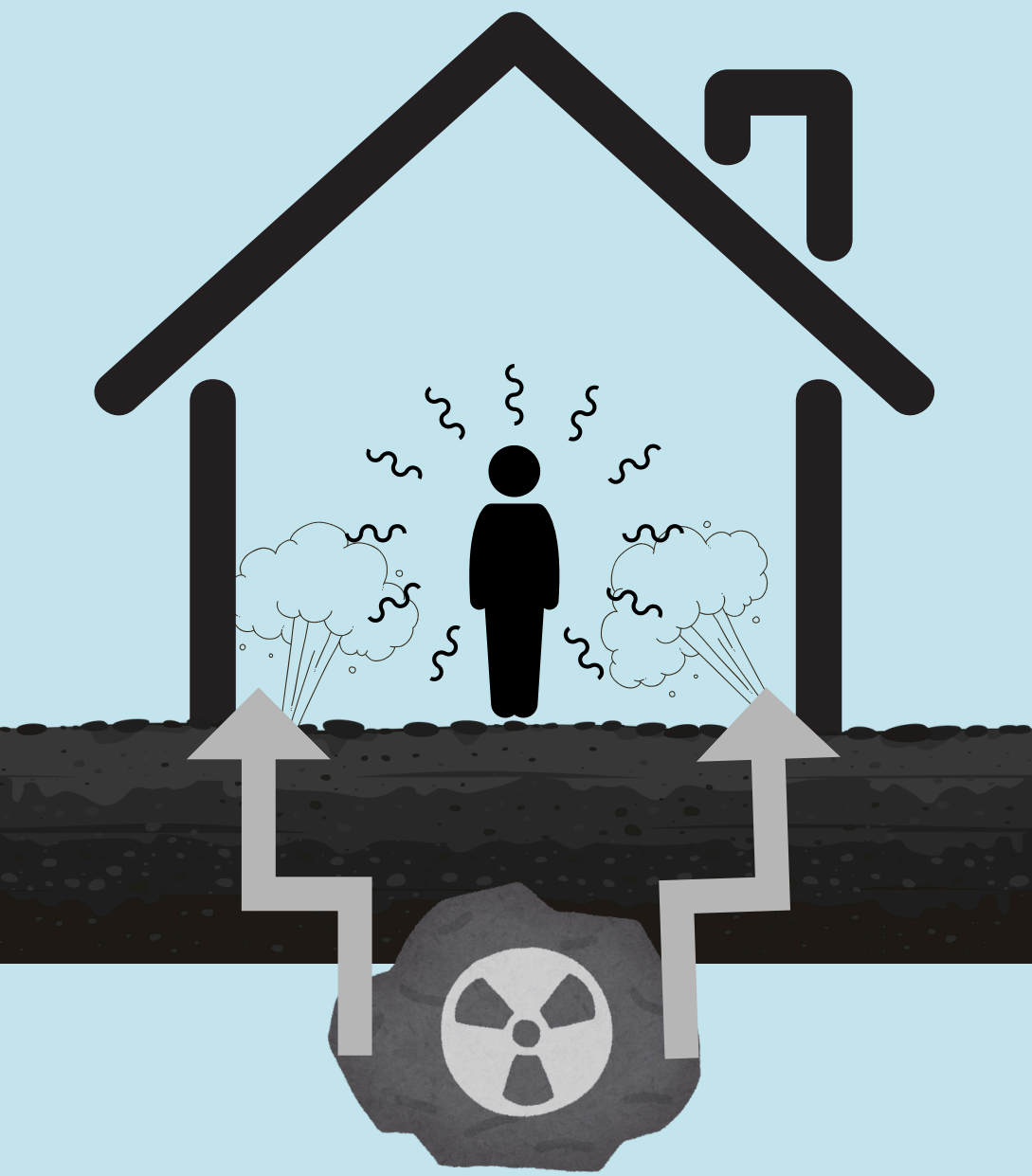
Where most of your exposure to radiation will come from. It occurs from natural sources found in the world

Artificially.

Where some of your exposure to radiation will come from. It occurs from practices to use radioactive material for development and medical reasons.



NATURAL



Radon and Thoron inhalation

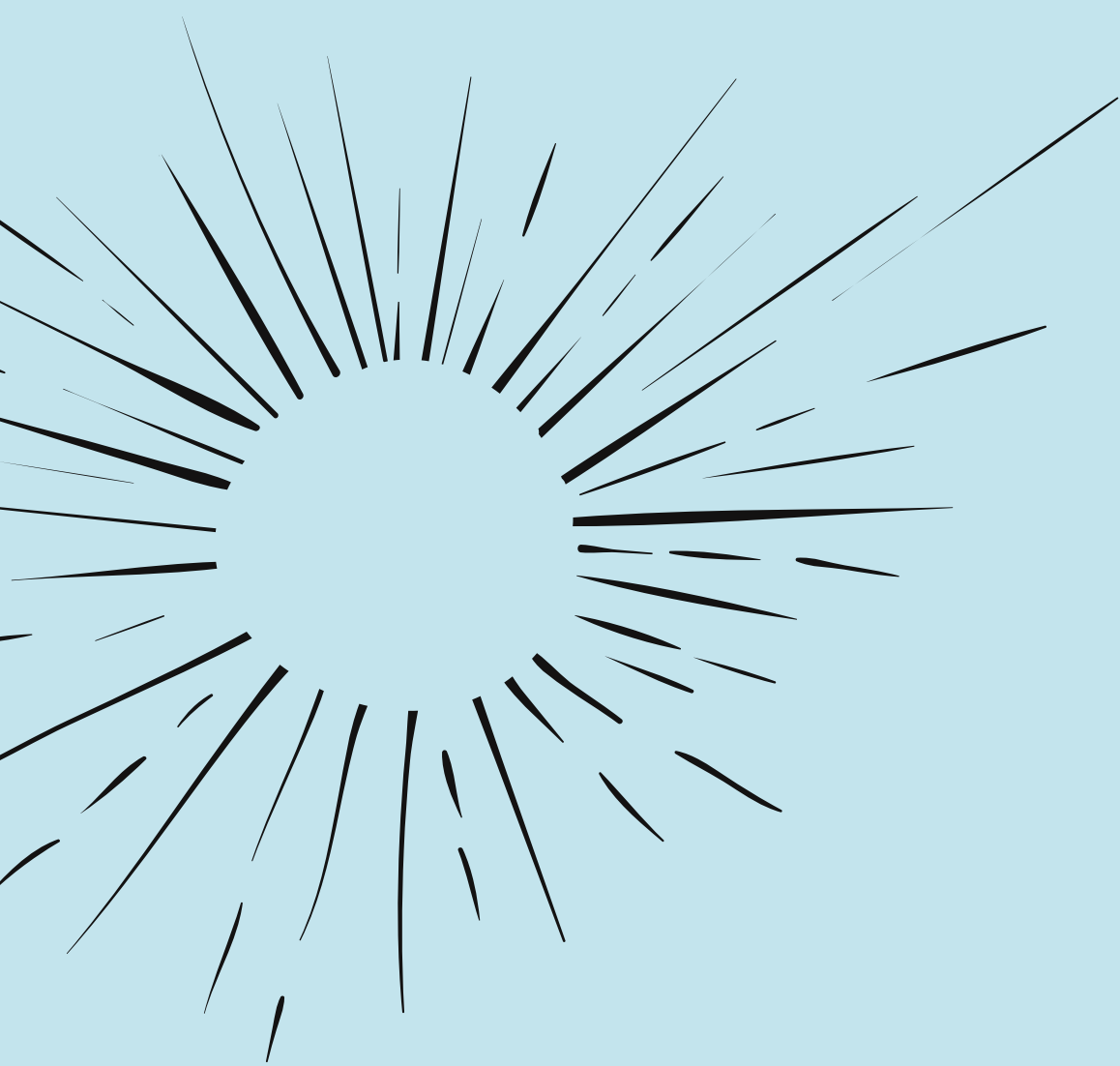
naturally occurring uranium or thorium slowly seeps through the ground and then gets inhaled

Uranium-238 ($\rightarrow$ radium-226 $\rightarrow$ radon-222)

Thorium-232 ($\rightarrow$ thoron)

Things that can influence this radiation are: construction/ventilation/foundation, amount of radioactive material in ground, and time spent inside.

roughly 228mrem/year



Cosmic Radiation

Rays are naturally penetrating the atmosphere from radioactive sources in space. Things like the sun, stars or cosmic events.

Carbon-14

Things that can influence this radiation are: Being at a higher altitude. Being closer to the poles. Flying and the amount of atmospheric shielding.

roughly 28mrem/year



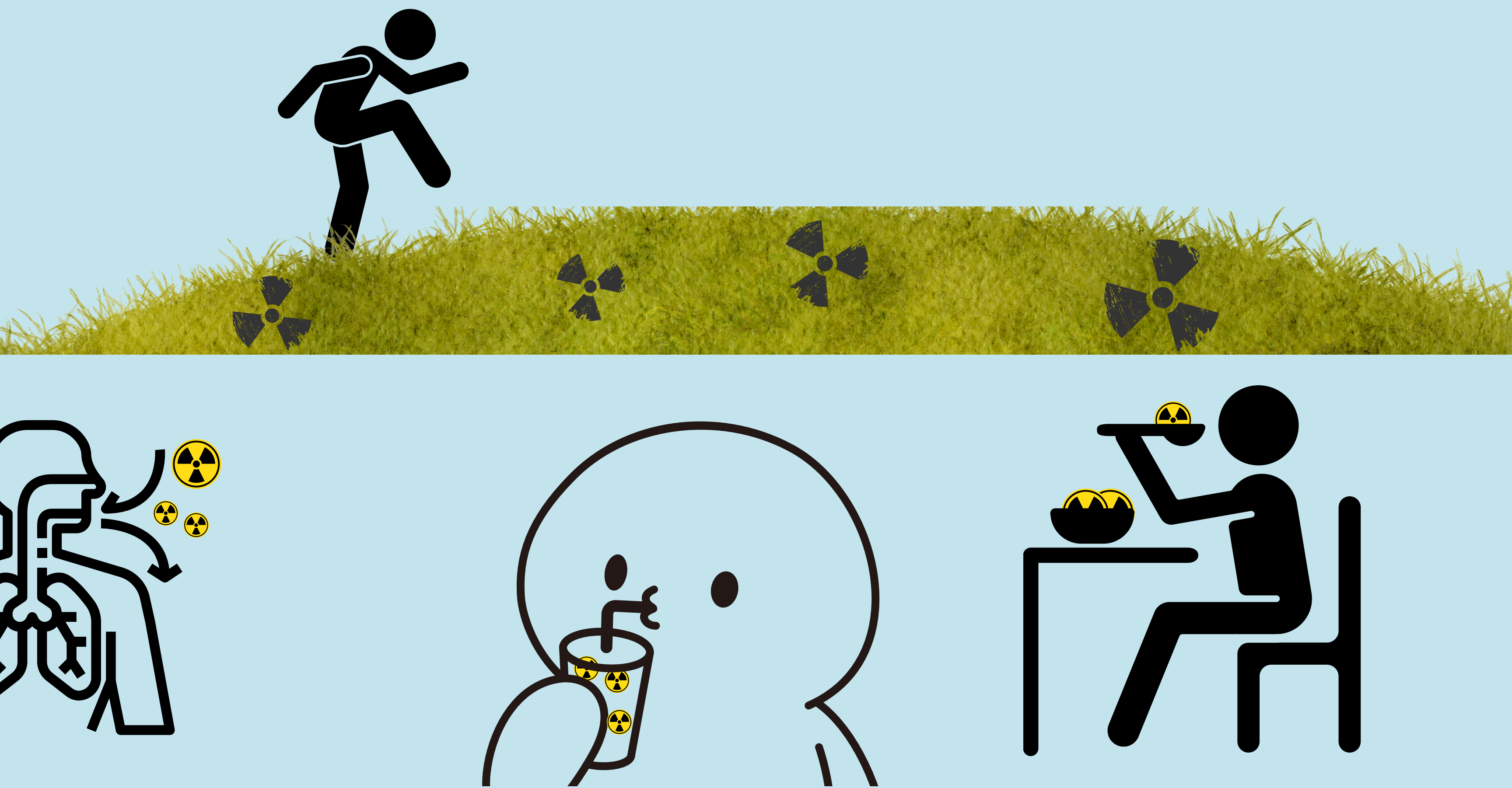
Terrestrial Radiaiton

There are minerals/material that occur naturally that are inherently radioactive. Such as uranium, thorium, potassium, etc. The ground is to a degree radioactive.

Uranium-238, Thorium-232, Potassium-40

Things that can influence this radiation are: The local geology, building material, soil composition and time spent outside.

roughly 21mrem/year



Internal Radiation

What we take in our bodies (eat, drink, breath) can be radioactive.

Potassium-40, Carbon-14, Radium-226

Things that can influence this radiation are: Diet, water composition, what's in the air, and physique.

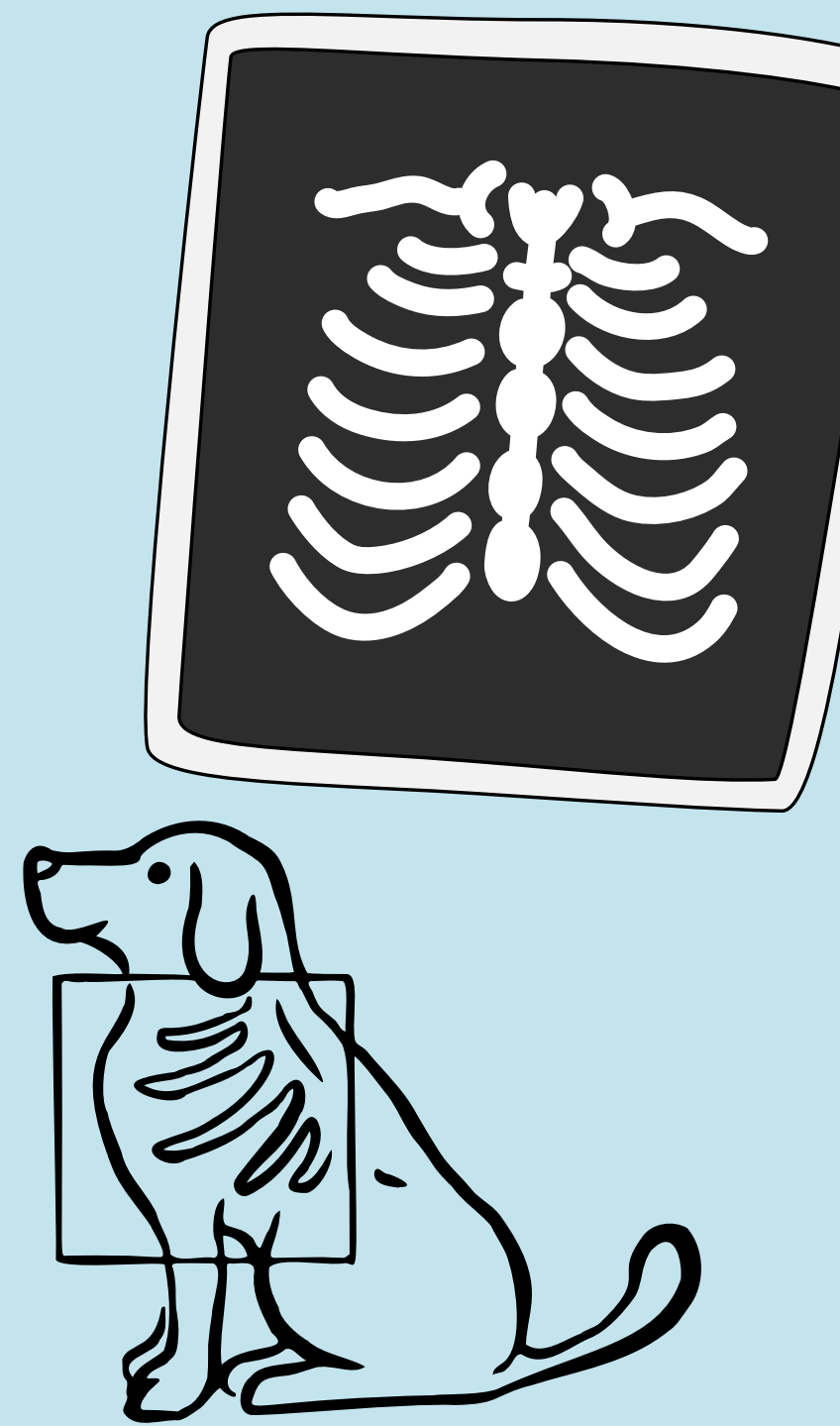
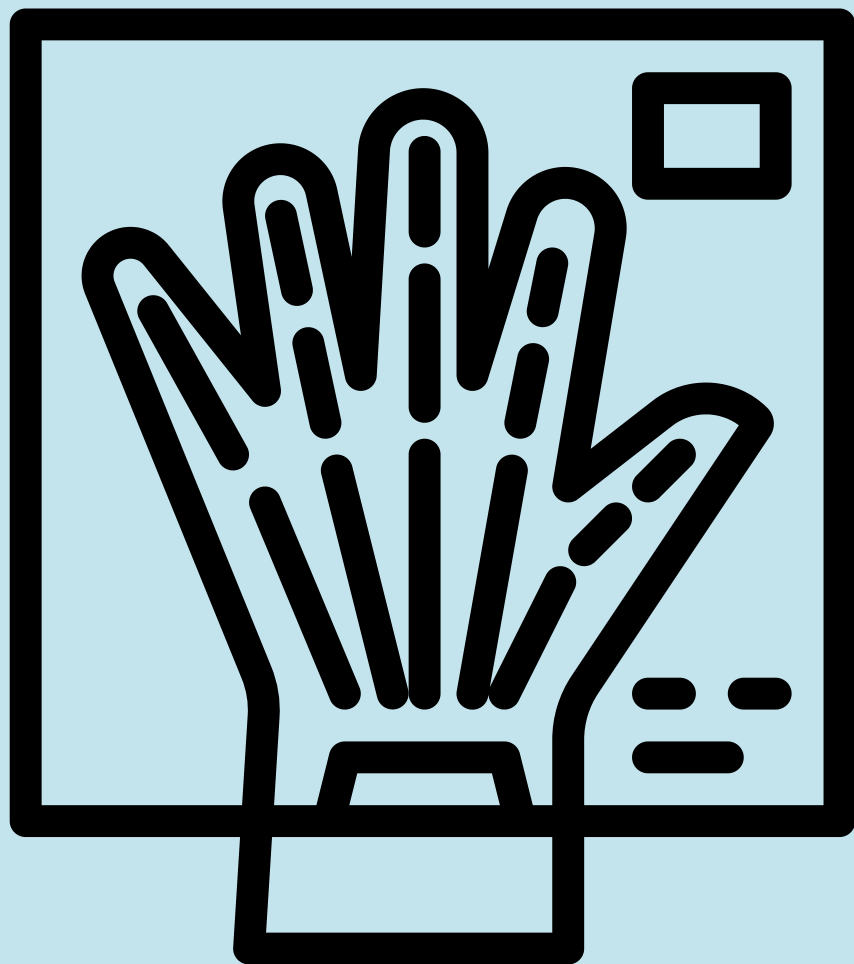
roughly 40mrem/year

ARTIFICIAL

Medical Imaging

When radiation is used to get a perspective of subdermal human body parts xray, ct, etc.

roughly 300mrem/year



Nuclear Power

When radiation is used to produce power generation. You can be exposed by simply living near these types of generators.

roughly .1mrem/year



Commercial Products

The things we use on a daily basis can cause radiation! Such as smoke detectors, some watches, and other commercial applications.



roughly 10mrem/year

Air Travel

The way we travel can cause us to be exposed to radiation! For instance a flight across the country exposes you to an extended amount of cosmic radiation.

roughly 3mrem/year

